Engineering Calculator Suite in Python

Python-Based Engineering Computation Toolkit

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Developed Using

Python
Visual Studio Code
Google Colab

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1 Project Overview

The Engineering Calculator Suite is a Python-based engineering application developed to perform common calculations encountered in mechanical and general engineering practice. The project combines multiple engineering calculators into a single modular software package, allowing users to perform calculations related to fluid mechanics, strength of materials, machine design, and hydraulic systems.

The primary objective of the project is to demonstrate the practical application of Python programming in solving engineering problems while emphasizing modular programming, user interaction, and computational analysis.

The software was developed as part of a Summer 2026 engineering skills development roadmap focused on Python programming and engineering computation.

2 OBJECTIVES

- Develop a modular engineering software application using Python.
- Implement commonly used engineering calculations.
- Apply engineering formulas through user-friendly programs.
- Practice function-based programming and code organization.
- Demonstrate engineering computation using Python.
- Create a portfolio-ready project for GitHub documentation.

3 CALCULATORS IMPLEMENTED

3.1 REYNOLDS NUMBER CALCULATOR

The Reynolds Number Calculator determines the flow regime of a fluid based on fluid properties and flow conditions.

Formula:

$$\mathrm{Re} = \frac{\rho \mathrm{VD}}{\mu}$$

Where:

- ρ = Fluid Density (kg/m^3)
- V = Fluid Velocity (m/s)
- D = Characteristic Diameter (m)
- μ = Dynamic Viscosity (Pa·s)

Output:

- Reynolds Number
- Flow Classification:
    - Laminar Flow
    - Transitional Flow
    - Turbulent Flow

3.2 BEAM STRESS CALCULATOR

The Beam Stress Calculator evaluates normal stress produced by an applied load.

Formula:

$$\sigma = \frac{\mathrm{F}}{\mathrm{A}}$$

Where:

- F = Applied Force (N)
- A = Cross-Sectional Area (m^2)

Output:

- Stress (Pa)

Applications:

- Structural Design
- Machine Components
- Material Testing

3.3 Factor of Safety Calculator

The Factor of Safety Calculator determines whether a component design is safe under a specified load.

Formula:

$$\mathrm{FoS} = \frac{\sigma_{\mathrm{Y}}}{\sigma}$$

Where:

- σ_{Y} = Yield Strength
- σ = Applied Stress

Output:

- Factor of Safety
- Safe or Unsafe Design Assessment

3.4 Spring Force Calculator

The Spring Force Calculator applies Hooke's Law to determine the force generated by a spring.

Formula:

$$\mathrm{F} = \mathrm{kx}$$

Where:

- k = Spring Constant (N/m)
- x = Spring Displacement (m)

The calculator also generates force-displacement data using NumPy and visualizes the relationship through a graph.

Applications:

- Suspension Systems
- Mechanical Springs
- Machine Design

3.5 Pump Power Calculator

The Pump Power Calculator determines hydraulic power requirements.

Formula:

$$P = \rho g Q H$$

Where:

- ρ = Fluid Density (kg/m^3)
- g = Gravitational Acceleration (9.81 m/s^2)
- Q = Flow Rate (m^3/s)
- H = Head (m)

Output:

- Hydraulic Power (W)

Applications:

- Fluid Transport Systems
- Pump Selection
- Hydraulic Engineering

4 PYTHON LIBRARIES USED

The project utilizes several Python libraries for engineering computation and visualization.

4.1 NUMPY

Used for:

- Numerical calculations
- Array generation
- Engineering datasets

4.2 MATPLOTLIB

Used for:

- Force-displacement plots
- Engineering visualization
- Graph generation

4.3 PANDAS

Used for:

- Experimental data processing
- CSV file handling
- Engineering data analysis

5 Example Outputs

5.1 Main Call

```
from reynolds_number import reynolds_calculator
from beam_stress import beam_stress_calculator
from factor_of_safety import fos_calculator
from spring_force import spring_force_calculator
from pump_power import calculate_pump_power

def main():
    while True:
        print("\nEngineering Calculator Suite")
        print("1. Reynolds Number Calculator")
        print("2. Beam Stress Calculator")
        print("3. Factor of Safety Calculator")
        print("4. Spring Force Calculator")
        print("5. Pump Power Calculator")
        print("6. Exit")

        choice = input("Choose Calculator: ")

        if choice == "1":
            reynolds_calculator()

        elif choice == "2":
            beam_stress_calculator()

        elif choice == "3":
            fos_calculator()

        elif choice == "4":
            spring_force_calculator()

        elif choice == "5":
            calculate_pump_power()

        elif choice == "6":
            print("Exiting Engineering Calculator Suite.")
            break

        else:
            print("Invalid choice. Please select 1 to 6.")

main()
```

Output:

```
Engineering Calculator Suite
1. Reynolds Number Calculator
2. Beam Stress Calculator
3. Factor of Safety Calculator
4. Spring Force Calculator
5. Pump Power Calculator
6. Exit
Choose Calculator:
```

5.2 Example 1: Reynolds Number Calculator

Input:

- Density = 1000 kg/m^3
- Velocity = 2 m/s
- Diameter = 0.05 m
- Viscosity = 0.001 Pa·s

Output:

- Re = 100000
- Flow Regime = Turbulent

```
Choose Calculator: 1
Density (kg/m^3): 1000
Velocity (m/s): 2
Diameter (m): 0.05
Viscosity (Pa.s): 0.001
Reynolds Number =  100000.0
Turbulent Flow
```

5.3 Example 2: Beam Stress Calculator

Input:

- Force = 5000 N
- Area = 0.01 m^2

Output:

- Stress = 500000 Pa

```
Choose Calculator: 2
Enter force: 5000
Enter area: 0.01
Stress =  500000.0
```

5.4 Example 3: Pump Power Calculator

Parameters:

- Gravitational Acceleration = 10 m/s^2
- Density = 1000 kg/m^3

Input:

- Flow Rate = 0.05 m^3/s
- Head = 10 m

Output:

- Power = 4905 W

```
Choose Calculator: 5
Enter the flow rate in cubic meters per second (m³/s): 0.05
Enter the head in meters (m): 10
The required pump power is: 4905.0 watts (W)
```

6 Engineering Applications

The Engineering Calculator Suite can be applied in multiple engineering disciplines.

Fluid Mechanics:

- Reynolds Number Analysis
- Pump System Design

Strength of Materials:

- Stress Evaluation
- Material Selection

Machine Design:

- Spring Force Calculations
- Factor of Safety Assessment

Engineering Education:

- Formula Verification
- Learning Engineering Concepts

The project demonstrates how programming can automate repetitive engineering calculations and improve analytical efficiency.

7 Future Improvements

Several enhancements can be implemented in future versions:

- Graphical User Interface (GUI)
- Unit Conversion Module
- Additional Thermodynamics Calculators
- Heat Transfer Calculators
- Numerical Methods Integration
- Data Export Features
- Engineering Database Support
- Error Handling Improvements

These improvements would transform the project from a calculator collection into a comprehensive engineering software toolkit.

8 Conclusion

The Engineering Calculator Suite successfully demonstrates the integration of Python programming with engineering analysis. The project combines multiple engineering calculations within a modular framework and provides a foundation for future development into larger engineering software applications.

The project strengthened skills in Python programming, modular software design, engineering computation, data analysis, and technical documentation while providing practical tools relevant to mechanical engineering applications.